

Sebastian Kopf

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1. Research Interests

Geomicrobiology, stable isotope geochemistry, microbial physiology, mass spectrometry

2. Education

2008 BS in Geosciences & Astrophysics, Jacobs University, Bremen, Germany

2010 MS in Geochemistry, MIT, Cambridge, MA

2014 PhD in Geobiology, California Institute of Technology, Pasadena, CA

3. Employment

2024-present Associate Professor with tenure, Department of Geological Sciences, CU Boulder

2024-2025 Visiting Professor, Geo- und Umweltforschungszentrum, Universität Tübingen, Germany

2016-2024 Assistant Professor, Department of Geological Sciences, CU Boulder

Parental leave semesters: Spring 2019, Spring 2021

2015-2016 Postdoctoral Research Fellow, Princeton University

4. Awards, and Honors

2025 Research Fellow, Alexander von Humboldt Foundation

2020 National Science Foundation Early Career Research Award

2015 Postdoctoral Fellowship Award, Agouron Institute

2014 Lawrence L. and Audrey W. Ferguson Prize for the outstanding doctoral thesis of the year, Division of Biology, California Institute of Technology

2011 International Student Predoctoral Fellowship Award, Howard Hughes Medical Institute

2008 Charles Vest Presidential Fellowship Award, Massachusetts Institute of Technology

5. Research and Creative Works

Key: members of the Kopf lab are highlighted in **bold**: **graduate student***, **undergraduate student****, **postdoc[#]**, **research staff[†]**. Informally advised students (e.g. as member of their thesis committee) are highlighted in *italic*: *student[§]*.

Peer-Reviewed Publications

Rosendahl[§] C.D., Zhao X., **Maloney[†] A.**, McFarlin J.M., Keller K.J., Kreutz R., Altun J., Bachmann L.H., Hartmann J., **Batther* H.K.**, Calhoun A.N., Elvert M., Pearson A., Leavitt W.D., **Kopf S.**, Elling F.J. Improved procedures for the cleavage and reduction of GDGTs for hydrogen isotope analysis. *Organic Geochemistry*, 2026, 105139. doi: 10.1016/j.orggeochem.2026.105139

Batther* H.K., Templeton AS, Hoehler T., Howells A., Bill M., Gropp J., **Kopf, S.** The stable carbon isotope fractionation of methanogenesis products at complete carbon consumption. *Geochemical Perspectives Letters*, 2025, 37, 35–39. doi: 10.7185/geochemlet.2543

Caro* T., McFarlin J. M., **Maloney[†] A. E.**, Jech S. D., Barker, A. J., Douglas, T. A., Barbato, R. A., **Kopf, S.** Microbial resuscitation and growth rates in deep permafrost: Lipid stable isotope probing results from the permafrost research tunnel in Fox, Alaska. *Journal of Geophysical Research: Biogeosciences*, 130, 2025, doi: 10.1029/2025JG008759

Keller K., **Kopf S.**, McFarlin J., **Maloney[†] A.**, Rhim J., Elling F., Ashing-Giwa K., Baker I., Calhoun A., Pearson A. Compound-specific carbon and hydrogen isotope analysis traces archaeal lipid signatures in cold seep marine systems. *Geochimica et Cosmochimica Acta*, 2025, doi: 10.1016/j.gca.2025.08.017

Mahapatra, S., **Kopf, S.H.**, **Halamka* T.**, Merten, C., Drost, D.A. and Minnaard, A.J. Total Synthesis of the Membrane Spanning Tetraether Lipid brGDGT Ia and Elucidation of Its Stereochemical Configuration. *Chem. Eur. J.*, 2025, doi: 10.1002/chem.202500702
Journal Cover Feature Article.

- Han[§] E., Kopf S., Maloney[†] A. E., Ai, X. E., Sigman, D. M., Zhang, X. Nitrogen Stable Isotope Fractionation by Biological Nitrogen Fixation Reveals Cellular Nitrogenase is Diffusion Limited. *PNAS Nexus*, 2025, Volume 4, Issue 3, doi: 10.1093/pnasnexus/pgaf061
- Harris[§] C., Kopf S., Rhim[#] J., Cobban A., Elling F., Feng X., McFarlin[#] F., Weber Y., Zhang Y., Zhou A., Batther* H.K., Pearson A., Leavitt W. Lipid hydrogen isotope compositions primarily reflect growth water in the model archaeon *Sulfolobus acidocaldarius*. *Applied and Environmental Microbiology*, 2025, 91:e01983-24, doi: 10.1128/aem.01983-24
- Asamoto* C.K., Ryu Y., Eckart K.N., Kelley-Kern** J., Dietrich L.E., Sigman D.M., Kopf S.H. Stable isotopic signature of dissimilatory nitrate reduction is robust against enzyme mutation. *PNAS*, 2024, 121 (48) e2416002121, 10.1073/pnas.2416002121
- Rhim[#] J., Kopf S., McFarlin[#] J., Maloney[†] A. E., Batther* H., Harris C. M., Zhou A., Feng X., Weber Y., Hoeft-McCann S., Pearson A., Leavitt W. D. Metabolic imprints in the hydrogen isotopes of *Archaeoglobus fulgidus* tetraether lipids. *Geochimica et Cosmochimica Acta*, 2024, 0016-7037. doi: 10.1016/j.gca.2024.09.032
- Harning D., Raberg J., McFarlin[#] J., Axford Y., Florian C., Ólafsdóttir K., Kopf S., Sepúlveda J., Miller G., and Geirsdóttir A. Spatiotemporal variation of modern lake, stream, and soil water isotopes in Iceland. *Hydrology and Earth System Sciences*, 2024; 28(18), 4275-4293. doi: 10.5194/hess-28-4275-2024
- Caro* T., Kashyap S.; Brown G., Chen C.; Kopf S.; Templeton A. Single-cell measurement of microbial growth rate with Raman microspectroscopy. *FEMS Microbiology Ecology*, 2024; fiae110, doi: 10.1093/femsec/fiae110
FEMS Microbiology Editor's Choice' article of the month.
- Halamka* T., Garcia A., Evans T., Schubert** S., Younkin[†], A. D., Hinrichs K-U., Kopf S. Occurrence of Ceramides in the *Acidobacterium Solibacter usitatus*: Implications for Bacterial Physiology and Sphingolipids in Soils. *Frontiers in Geochemistry*, Vol 2, 2024. doi: 10.3389/fgeoc.2024.1400278
- Shipley O., Anna J., Gabriel J., Hayden B., Pauli J., Jordan C., Anderson L., Bailey A., Bataille C., Cicero C., Close H., Cook C., Cook J., Desai A., Evaristo J., Filley T., France C., Kim S., Kopf S., Loisel J., Manlick P., McFarlin[#] J., McMeans B., O'Connell T., Semmens B., Stantis C., Szejner P., Pilaar Birch S. Putman A., Stricker C., Trammell, T., Uhen M., Weintraub-Leff S., Wooler M., Willimas J., Yarnes C., Vander Zanden H., Newsome S.: Design, development, and implementation of IsoBank: a centralized repository for isotopic data. *PLoS One*. 2024 Sep 6;19(9):e0295662. doi: 10.1371/journal.pone.0295662.
- Maloney[†] A. E., Kopf S., Zhang Z., McFarlin[#] J., Nelson D. B., Masterson A. L., Zhang X. Large enrichments in fatty acid ²H/¹H ratios distinguish respiration from aerobic fermentation in yeast *Saccharomyces cerevisiae*. *PNAS*, 121 (20) e2310771121, 2024. doi: 10.1073/pnas.2310771121
- Kantnerová[#] K., Kuhlbusch N., Juchelka D., Hilkert A., Kopf S., Neubauer[†] C.: ESI-Orbitrap IRMS: A Guide to Precision Isotopocule Analysis for Life and Environmental Scientists. *Nat Protoc*. 2024 Aug;19(8):2435-2466. doi: 10.1038/s41596-024-00981-5
- Havranek[§], R.; Snell, K.; Kopf, S.; Davidheiser-Kroll, B.; Claymore, V.; Vaugh, B. Technical note: Lessons from and best practices for the deployment of the Soil Water Isotope Storage System. *Hydrology and Earth System Sciences*, 27: 2951-2971, 2023. doi: 10.5194/hess-27-2951-2023
- Leavitt, W.; Kopf S.; Weber, Y.; Chiu, B.; McFarlin[#], J. M.; Zhou, A.; Zhang, Y.; Elling, F.; Hoeft-McCann, S.; Pearson, A. Controls on the hydrogen isotope composition of tetraether lipids in an autotrophic ammonia-oxidizing marine archaeon. *Geochimica et Cosmochimica Acta*, 352: 194-210, 2023. doi: 10.1016/j.gca.2023.04.033
- Caro*, T. A.; McFarlin[#], J. M.; Jech[§], S.; Fierer, N; Kopf, S. Hydrogen stable isotope probing of lipids demonstrates slow rates of microbial growth in soil. *Proceedings of the National Academy of Sciences*, 120:16, 2023. doi: 10.1073/pnas.2211625120
- Rempfert[§], K. R.; Nothhaft[§], D. B.; Kraus, E. A.; Asamoto*, C. K.; Evans, R. D.; Spear, J. R.; Matter, J. M.; Kopf, S. H.; Templeton, A. S. Subsurface biogeochemical cycling of nitrogen in the actively serpentinizing Samail Ophiolite, Oman. *Frontiers in Microbiology*, 2023. doi: 10.3389/fmicb.2023.1139633

- Boettger, J, **Neubauer**[†], **C, Kopf, S**, and Kubicki, J. Microbial Denitrification: Active Site and Reaction Path Models Predict New Isotopic Fingerprints. *ACS Earth and Space Chemistry*, 6(11): 2582-2594, 2022, doi: 10.1021/acsearthspacechem.2c00102
- Halamka***, **T. A.**; *Raberg*[§], *J. H.*; **McFarlin**[#], **J. M.**; **Younkin**[†], **A. D.**; **Mulligan**^{**}, **C.**; Liu, X.; and **Kopf, S. H.** Production of diverse brGDGTs by *Acidobacterium Solibacter usitatus* in response to temperature, pH, and O₂ provides a culturing perspective on brGDGT proxies and biosynthesis. *Geobiology*, 2022, doi: 10.1111/gbi.12525
- Halamka***, **T A**; **McFarlin**[#], **J M**; **Younkin**[†], **A D**; **Depoy**^{**}, **J**; Dildar, N; and **Kopf, S H.** Oxygen limitation can trigger the production of branched GDGTs in culture. *Geochemical Perspectives Letters*, 19: 36–39, 2021, doi: 10.7185/geochemlet.2132
- Fripiat F, Martínez-García A, Marconi D, Fawcett S, **Kopf S**, Luu V, Rafter P, Zhang R, Sigman D, Haug G. Nitrogen isotopic constraints on nutrient transport to the upper ocean. *Nature Geoscience*, 2021, doi: 10.1038/s41561-021-00836-8
- Nothhaft*[§], D, Templeton, A, Rhim, J, Wang, D, Labidi, J, Miller, H, Boyd, E, Matter, J, Ono, S, Young, E, **Kopf, S**, Kelemen, P, Conrad, M, and Team T. O. D. P. S. Geochemical, Biological, and Clumped Isotopologue Evidence for Substantial Microbial Methane Production Under Carbon Limitation in Serpentinites of the Samail Ophiolite, Oman. *Journal of Geophysical Research: Biogeosciences*, 126(10), 2021, doi: 10.1029/2020JG006025
- Boudinot*[§] **G, Kopf S**, Dildar N, Sepulveda J. Carbon Cycling During Oceanic Anoxic Event 2: Compound-Specific Carbon Isotope Evidence From the Western Interior Seaway. *Paleoceanography and Paleoclimatology*, 36(9): e2021PA004287, 2021, doi: 10.1029/2021PA004287
- Hilkert, A.; Böhlke, J. K.; Mroczkowski, S. J.; Fort, K. L.; Aizikov, K.; Wang, X. T.; **Kopf, S. H.**; and **Neubauer, C.** Exploring the Potential of Electrospray-Orbitrap for Stable Isotope Analysis Using Nitrate as a Model. *Analytical Chemistry*, 93(26): 9139-9148, 2021, doi: 10.1021/acs.analchem.1c00944
- Kopf, S.**; Davidheiser-Kroll, B.; and Kocken, I. Isoreader: An R package to read stable isotope data files for reproducible research. *Journal of Open Source Software*, 6(61): 2878. 2021, doi: 10.5194/bg-18-3579-2021
- Asamoto C**, *Rempfert*[§] **K**, Luu V, **Younkin**[†] **A**, Sigman D, **Kopf S**. Enzyme-Specific Coupling of Oxygen and Nitrogen Isotope Fractionation of the Nap and Nar Nitrate Reductases. *Environmental Science & Technology*, 2021, 55(8): 5537-5546, doi: 10.1021/acs.est.0c07816
- Lengger S, Weber Y, Tayler K, **Kopf S**, Berstan R, Bull I, Mayser J-P, Leavitt W, Blewett J, Pearson A, Pancost R. Determination of the $\delta^2\text{H}$ values of high molecular weight lipids by high temperature GC coupled to isotope ratio mass spectrometry. *Lengger et al. Rapid Communications in Mass Spectrometry* (2021), doi: 10.1002/rcm.8983
- Bhatnagar S, **Cowley**^{**} **E**, **Kopf S**, Pérez Castro S, Kearney S, Dawson S, Hanselmann K, Ruff SE. Microbial community dynamics and coexistence in a sulfide-driven phototrophic bloom. *Environmental Microbiome* 15, 3 (2020) doi:10.1186/s40793-019-0348-0
- Silverman**^{**} **S**, **Kopf S**, Gordon R, Bebout B, Som S. Morphological and isotopic changes of heterocystous cyanobacteria in response to N₂ partial pressure. *Geobiology*. 2019, 17(1), 60-75, doi: 10.1111/gbi.12312
- McRose D, Lee A, **Kopf S**, Baars O, Kraepiel A, Sigman D, Morel F, Zhang X. Effect of Iron Limitation on the Isotopic Composition of Cellular and Released Fixed Nitrogen in *Azotobacter vinelandii*. *Geochimica Et Cosmochimica Acta*, 2019, 244, doi: 10.1016/j.gca.2018.09.023
- Neubauer C, Sessions A, Booth I, Bowen B, **Kopf S**, Newman D, Dalleska N. Towards measuring growth rates of pathogens during infections by D₂O-labeling lipidomics. *Rapid Communications in Mass Spectrometry*, 2018, 32, 2129-2140, doi: 10.1002/rcm.8288
- Neubauer C, Kasi A, Grahl N, Sessions A, **Kopf S**, Kato R, Hogan D, Newman D. Refining the application of microbial lipids as tracers of *Staphylococcus aureus* growth rates in cystic fibrosis sputum. *Journal of Bacteriology*, 2018, 200 (24), doi: 10.1128/JB.00365-18
- Miller*[§] **H**, Chaudhry N, Conrad M, Bill M, **Kopf S**, Templeton A. Large carbon isotope variability during alkaline methanogenesis. *Geochimica Et Cosmochimica Acta*, 2018, 237, 18-31, doi: 10.1016/j.gca.2018.06.007

- Marconi^S D, **Kopf S**, Rafter P, Sigman D. Aerobic respiration along isopycnals leads to overestimation of the isotope effect of denitrification in the ocean water column. *Geochimica et Cosmochimica Acta*, 2017, 197, 417-432, doi: 10.1016/j.gca.2016.10.012
- Pasulka A, Thamtrakoln K, **Kopf S**, Guan Y, Poulos B, Moradian A, Sweredoski M, Hess S, Sullivan M, Bidle K, Orphan V. BONCAT and nanoSIMS-based methods for the direct measurement of newly synthesized viral particles and the flow of carbon and nitrogen after host lysis. *Environmental Microbiology*, 2017, 20(2), 671-692, doi: 10.1111/1462-2920.13996
- Kopf S**, Sessions AL, Cowley ES, Reyes C, Van Sambeek L, Hu Y, Orphan VJ, Kato R, Newman DK. Trace incorporation of heavy water reveals slow and heterogeneous pathogen growth rates in cystic fibrosis sputum. *PNAS*, 2016, 113 (2), doi: 10.1073/pnas.1512057112
- Kopf S**, McGlynn S, Green-Saxena A, Guan Y, Newman DK, Orphan VJ. Heavy water and 15N labelling with NanoSIMS analysis reveals growth rate dependent metabolic heterogeneity in chemostats. *Environmental Microbiology*, 2015, 17 (7), doi: 10.1111/1462-2920.12752
- Kramer D, **Kopf S**, Bau M. Oxidative mobilization of cerium and uranium and enhanced release of “immobile” high field strength elements from igneous rocks in the presence of the biogenic siderophore desferrioxamine B. *Geochimica Cosmochimica Acta*, 2015, 165, 263-279, doi: 10.1016/j.gca.2015.05.046
- Cowley ES, **Kopf S**, LaRiviere A, Ziebis W, Newman DK. Pediatric cystic fibrosis sputum can be chemically dynamic, anoxic and extremely reduced due to hydrogen sulfide formation. *mBio*, 2015, 6 (4), e00767-15, doi: 10.1128/mBio.00767-15
- Kopf S**, Henny C and Newman DK. Ligand-enhanced abiotic iron oxidation and the effects of chemical versus biological iron cycling in anoxic environments. *Environmental Science & Technology*, 2013, 47(6), 2602-2611, doi: 10.1021/es3049459
- Kopf S** and Ono S. Sulfur mass-independent fractionation in liquid phase chemistry: UV photolysis of phenacylphenylsulfone as a case study. *Geochimica Et Cosmochimica Acta*, 2012, 85, 160-169, doi: 10.1016/j.gca.2012.02.008
- Kopf S** and Newman DK. Photomixotrophic growth of *Rhodobacter capsulatus* SB1003 on ferrous iron. *Geobiology*, 2011, 10(3), 216-222, doi: 10.1111/j.1472-4669.2011.00313.x
- Kopf S**, Hallegatte S, Ha-Duong M. Using maps of city analogues to display and interpret climate change scenarios and their uncertainty. *Natural Hazards and Earth System Sciences*, 2008, 8 (4), 905-918, www.nat-hazards-earth-syst-sci.net/8/905/2008

Non-Peer-Reviewed Publications

- Bose A, **Kopf S**, Newman DK. From geocycles to genomes and back. In: *Microbial metal and metalloid metabolism: advances and applications* (Stolz JF & Oremland RS, Eds). ASM Press, 2010.

6. Grants (at CU Boulder)

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|-----------|---|
| 2026-2029 | DoD Strategic Environmental Research and Development Program (RC26-C1-0371). Kopf (PI at CU Boulder), McFarlin (lead PI, University of Wyoming), Doering, Shuman. <i>Developing Robust Long-Term Baselines of Wildfire, Ecosystem Variability and Transformation, and Their Drivers in the Donnelly Training Area, Alaska</i> . \$948,917 |
| 2025-2026 | DoD Defense University Research Instrumentation Program (W911NF-25-1-0155). Kopf (PI), Neubauer (Co-I). <i>Enabling transformative research at CU Boulder with an Orbitrap liquid chromatography (LC) isotope ratio mass spectrometry (IRMS) system</i> . \$513,947. |
| 2025-2026 | NASA Exobiology (80NSSC25K7765). Kopf (PI), Neubauer (Co-I). <i>Enabling transformative research with a versatile liquid chromatography front-end for an Orbitrap isotope ratio mass spectrometer</i> . \$48,047. |
| 2024-2027 | NSF Geoinformatics (2411458). Kopf (PI), Neubauer (Co-I, INSTAAR), Newsome (Co-PI, University of New Mexico), Yarnes (Co-PI, University of California Davis). <i>Collaborative</i> |

- Research: Innovative Resources: Into the ISOVERSE - open-source data tools for efficient, transparent, and reproducible processing of stable isotope data.* \$425,030 to CU Boulder.
- 2022-2025 NASA Exobiology (20NSSC22K1895). Templeton, Kopf (Co-Is at CU Boulder), Hoehler (PI at NASA Ames / lead institution). *Methanogenic Activity and Isotopic Biosignatures Under Carbon Limitation.* \$498,279 to CU Boulder.
- 2021-2024 Army Research Office Life Sciences Branch (W911NF2120119). Kopf (PI). *Microbial activity in a changing Arctic.* \$357,002 to CU Boulder / total.
- 2020-2025 NSF EAR Geobiology & Low Temperature Geochemistry (1945484). Kopf (PI). *CAREER: Uncovering the origins and biological purpose of the orphaned branched GDGT temperature biomarker.* \$537,699 to CU Boulder / total.
- 2020-2021 NSF OCE Chemical Oceanography (2041539). Kopf (PI), Neubauer (Co-I, INSTAAR). *EAGER: Unlocking the clumped isotope signatures of marine nitrate.* \$185,920 to CU Boulder / total.
- 2019-2022 NSF EAR Geobiology and Low Temperature Geochemistry (1928303). Kopf (PI at CU Boulder), Leavitt (PI at Dartmouth). *Collaborative Research: Establishing the hydrogen isotopic window into Archaeal lipid biomarkers.* \$280,180 to CU Boulder, \$555,115 total.
- 2019-2020 CU Boulder RIO SEED Grant. Kopf (PI), Fierer (Co-I, EBio), Wieder (Co-I, NCAR). *Measuring the pulse of global change at the interface of Earth & Life.* \$49,990 total (CU internal).
- 2017-2020 NASA Exobiology (80NSSC17K0667). Kopf (PI at CU Boulder), Zhang (PI at Princeton / lead institution), Kraepiel (CoI, Princeton). *What Controls the Nitrogen Isotope Fractionation of N₂ Fixation?* \$119,613 to CU Boulder, \$569,419 total.

7. Invited Presentations (since at CU Boulder)

Conferences and Seminars

- 2025 Geo. Sci. Seminar, University of Chicago, Chicago, IL.
- 2025 Chemistry & Biochemistry Seminar, University of Denver, Denver, CO.
- 2025 Seminar, Max Planck Institute for Chemistry, Mainz, Germany.
- 2024 Current Topics in Geosciences Colloquium, Universität Basel, Switzerland.
- 2024 GeoEnviron Seminar, University of Tübingen, Germany.
- 2024 Geobiology Gordon Research Conference, Ventura, CA.
- 2023 Geological Sciences Colloquium, University of Colorado, Boulder, CO.
- 2023 2nd International GDGT Workshop. ETH Zurich, Switzerland.
- 2022 Organic Geochemistry Gordon Research Conference. Holderness, NH.
- 2022 American Chemical Society Spring Meeting. San Diego, CA.
- 2022 Virtual Biological and Biotechnological Sciences Seminar, Army Research Laboratories.
- 2021 Earth Science Department Seminar, Dartmouth College, Hanover, NH.
- 2021 Society of Fellows, Dartmouth College, Hanover, NH.
- 2020 GSA Annual Meeting. Virtual.
- 2019 Geobiology Society Meeting. Banff, Canada.
- 2019 INSTAAR Seminar, University of Colorado Boulder, Boulder, CO.
- 2017 Lamont Doherty Earth Observatory Colloquium, Columbia University, NY.
- 2017 USGS Colloquium, Federal Center, Denver, CO.
- 2017 Civil & Environmental Engineering Seminar, Colorado School of Mines, Golden, CO.
- 2016 Telluride Science Research Conference on Iron Biogeochemistry, Telluride, CO.
- 2016 Geography and Environmental Engineering Colloquium, John's Hopkins University, Baltimore, MD.

Workshops and Summer Courses

- 21-24 Isocamp Summer Course on Stable Isotope Biogeochemistry & Ecology, University of New Mexico.
- 2019 IsoBank workshop at the University of New Mexico.
- 2019 Isocamp Summer Course on Stable Isotope Biogeochemistry & Ecology, University of Utah.

2016 International Geobiology Summer Course at the Wrigley Institute/University of Southern California/Colorado School of Mines.

8. Teaching (at CU Boulder)

GEOL 1180: Our Microbial Planet (Fall 2018, Spring 2020, Spring 2023, Spring 24)

New general interest science class on the crucial role microbes play on Earth - class open to all majors.

GEOL/MCDB 3181: Microbial Planet Lab (Fall 2022, Fall 2023)

New upper division hands-on research class isolating and characterizing soil microorganism – first course based undergraduate research experience (CURE) class in the department.

GEOL/MCDB/ENVS 4185 / GEOL 5185: Geomicrobiology (Fall 2017, Fall 2019, Spring 2022, Fall 2025)

New upper division flipped classroom using team-based learning to gain a quantitative understanding of microbial energetics and earth-life interactions.

GEOL/EBIO 4444 / 5444: Data Visualization (Fall 2021, Spring 2023, Spring 2026)

New graduate/undergraduate hands-on data analysis and visualization using tidyverse R.

GEOL 5042: Computational Tools (Fall 2018, Fall 2020)

New graduate course on advanced computational tools in R, python and microcontroller programming.

GEOL 5253: Stable Isotope Fractionation (Fall 2016, Spring 2018, Fall 2020)

New graduate class on stable isotope fractionation including modelling advanced isotope systematics.

9. Mentoring (since at CU Boulder)

Postdoctoral Researchers

2022-2024 Dr. Kristýna Kantnerová (co-advised with C. Neubauer)

2020-2023 Dr. Jeemin Rhim (co-advised with W. Leavitt at Dartmouth College)

2019-2022 Dr. Jamie McFarlin

Research Staff

2023-present Dr. Ashley Maloney

2020-present Dr. Cajetan Neubauer

2018-present Adam Younkin

2016-2021 Dr. Nadia Dildar

Graduate Students

2025-present GEOL/IQ Bio PhD student Alana Robinson

2023-present GEOL/IQ Bio PhD student Catherine Fontana (co-advised with B. Wing)

2020-2025 GEOL PhD student Harp Batther (co-advised with A. Templeton)

2020-2024 GEOL/IQ Bio PhD student Tristan Caro (co-advised with A. Templeton)

2019-2023 GEOL PhD student Toby Halamka

2017-2023 GEOL PhD student Ciara Asamoto

2019/2020 IQ Biology rotation student Tristan Caro

2018/2019 IQ Biology rotation student Corinne Walsh

2016/2017 IQ Biology rotation student Kelsey Dahlgren

2016/2017 IQ Biology rotation student Elise Cowley

Undergraduate Students

Fall 2023 UROP student Carolyn Bivins

Summer 2022 RECCS summer student Madison Scarlotti

2022-2023 UROP student Madigan Rumley

Summer 2022 RECCS summer student Adeena Chughtai

Summer 2022 RECCS summer student Sal Greenberger

2022-2023 UROP student Julia Kelley-Kern

2021-2022 UROP student Stephanie Schubert

Summer 2021 Summer research assistant Josie Mottram

Summer 2021 Summer research assistant Christopher Mulligan
2019-2020 UROP student Joely Depoy
Summer 2019 RECCS summer student Alec Elder
2016-2017 MCDB Honors thesis student Shaelyn Silverman

Training Visitors

Summer 2024 Amanda Calhoun, PhD student, Havard University
Summer 2024 Vivien Schneider, PhD student, University of Bristol
Summer 2024 Chris Rosendahl, PhD student, University of Kiel
Spring 2023 Olivia Pendas, undergraduate student, Dartmouth College
Summer 2022 Alta Howells, Postdoc, NASA Ames
Fall 2021&23 Carolyn Harris, PhD student, Dartmouth College
Spring 2019 Yuki Weber, Postdoc, Harvard University
Summer 2018 Ashley Maloney, Postdoc, Princeton University
Summer 2018 Eunah Han, PhD student, Princeton University
Summer 2017 Matthew Salie, Postdoc, Scripps Research Institute

Graduate Student Committees

2026-present EBIO PhD student Douglas Castro
2025-present GEOL PhD student Mobeen Zahid
2025-present EBIO PhD student Asher Marvey
2023-present GEOL PhD student Harry Allbrook
2022-present EBIO PhD student Jessica Rush
2022-present GEOL PhD student Kevin Rozmiarek
2023-2025 EBIO PhD student Laura Rea
2022-2025 Dartmouth College PhD student Carolyn Harris
2021-2024 EBIO PhD student Sierra Jech
2018-2024 Princeton PhD student Eunah Han
2017-2024 GEOL PhD student Ellie Hara
2017-2023 GEOL PhD student Jennifer Reeve
2019-2023 Colorado School of Mines PhD student Patrick Thieringer
2018-2023 EBIO PhD student Corinne Walsh
2021-2022 GEOL PhD student Rachel Havranek
2021-2022 Biochemistry PhD student Kelsey Dahlgreen
2017-2022 GEOL PhD student Sebastian Cantarero
2020-2021 EBIO MSc student Molly Huber
2019-2021 GEOL PhD student Katie Rempfert
2018-2021 EBIO MSc student Amy Barfield
2017-2021 GEOL PhD student Jonathan Raberg
2016-2021 GEOL PhD student Mike Zawaski
2017-2020 INSTAAR/CEAE PhD student Anna Bergstrom
2017-2020 GEOL PhD student Daniel Nothaft
2017-2020 GEOL PhD student Garrett Boudinot
2017-2019 GEOL PhD student Lina Perez Angel
2016-2018 GEOL/USGS MSc student Tyler Kane

10. Professional Development (since at CU Boulder)

2023 CU Boulder Inclusive Mentoring for Research Experience Mentors
2021 CU Boulder Supporting Student Resiliency Series
2020 Small World Initiative Workshop: Designing and implementing Course-based Undergraduate Research Experience (CURE) lab classes.

- 2018 CU Boulder PI Academy Workshop: Navigating Resources for Research, Scholarship and Creative Work
- 2018 CU Boulder PI Academy Workshop: Research and Seed Grant Mixer
- 2018 CU Boulder PI Academy Workshop: Packard & Keck Fellowships Workshop
- 2018 CU Boulder FTEP Workshop: Flipping the classroom
- 2017 CU Boulder FTEP Workshop: Student pushback in active classrooms
- 2017 Goldschmidt Workshop: Team-based-learning (TBL) for flipping Earth Sciences classrooms
- 2017 CU Boulder LEAP Workshop: Introductory Leadership
- 2016 NAGT/SERC Early Career Geoscience Workshop for Teaching, Research and Career Management

11. Service & Outreach (since at CU Boulder)

Department

- 2026-present Associate Chair for Undergraduate Studies
- 2025-present Graduate Curriculum Committees
- 2025-present Graduate Admission Committees
- 2022-2026 Undergraduate Curriculum Committees
- Fall 2022 Department Chair Search Committee
- 2021/2022 Graduate Curriculum Committees
- 2020/2021 Diversity, Equity & Inclusivity Committee
- 2019/2020 Executive Committee
- Fall 2018 Graduate Curriculum Committee
- Fall 2017 Committee of the Future
- 2017-2018 Web Committee (Chair)
- 2016/17 Space Committee

University

- 2019-2021 ASSETT Inclusive Data Science Innovation Incubator
- 2017-2018 RIO Campus Computing Resources Workgroup
- 2017 IQBiology Admissions Committee
- 2016 RIO Advisory Board Subcommittee on Engaging Faculty in Innovation

Community

- 2016-present Developer of open-source software and hardware for the geochemistry and geomicrobiology research communities including www.isoverse.org
- 2016-present Manuscript Reviewer for: Geobiology, Environmental Microbiology, Geochimica et Cosmochimica Acta, Nature Communications, Microbial Ecology, mBio, PLOS One, Environmental Microbiology, Chemical Geology, PNAS, Environmental Science & Technology
- 2016-present Proposal Reviewer for: NASA Astrobiology, NSF Geobiology and Low Temperature Geochemistry, NSF Chemical Oceanography, Department of Energy Office of Science, American Chemical Society Petroleum Research Fund, Novo Nordisk Foundation
- 2022 Proposal review panelist for NSF EAR Geobiology & Low Temperature Geochemistry program
- 2020-2021 Lead organizer of the CU Boulder visit days for community college students from the Denver Metro STEM Alliance for Minority Participation (comprising Community College of Aurora, Arapahoe Community College and Front Range Community College).
- 2020 Lead organizer for the first Isoverse Workshop on reproducible stable isotope data processing, University of Colorado Boulder, CO.
- 2019 Lead faculty organizer for the 3rd Rocky Mountain Geobiology Symposium, University of Colorado Boulder, CO.
- 2018 Co-Chair for AGU Conference Session: “Understanding the Biogeochemistry of Nitrogen Inputs and Outputs from Molecular to Global Scales”

- 2018 Co-Chair for Goldschmidt Conference Session: “Traditional and Non-Traditional Stable Isotopes in Geobiology and Biogeochemistry”
- 2017 Co-Chair for Goldschmidt Conference Session: “Novel tools and approaches for investigating biogeochemical cycling and organic signatures of microbial activity”
- 2016 Co-Instructor for GSA Short course: “Organic, Light and Clumped Stable Isotope Geochemistry in the 21st Century: Principles, Practices and Novel Applications” with K. Snell, J. Sepulveda, and B. Davidheiser-Kroll

12. Other Conference Presentations (since at CU Boulder)

- 2025 Join European Stable Isotopes Users group meeting, 1st inaugural International Orbitrap-IRMS User Meeting, Goldschmidt Conference, International Meeting on Organic Chemistry, Rocky Mountain Geobiology Symposium
- 2024 Geobiology Gordon Research Conference, American Chemical Society Spring Meeting, Rocky Mountain Geobiology Symposium, International Conference on Permafrost, Ecological Society of America Annual Meeting, Organic Geochemistry Gordon Research Conference, Goldschmidt Conference
- 2023 American Chemical Society Spring Meeting, Rocky Mountain Geobiology Symposium, Front Range Microbiome Symposium, American Society for Mass Spectrometry Annual Conference, Goldschmidt Conference, Advances in Stable Isotope Techniques & Applications (ASITA) Conference, Archaea Gordon Research Conference, 2nd International GDGT workshop, International Meeting on Organic Geochemistry
- 2022 Northeast Geobiology Symposium, Astrobiology Science Conference (AbSciCon), Goldschmidt Conference, International Symposium on Microbial Ecology (ISME), International Workshop on Microbial Life (Microenergy), International Clumped Isotope Workshop, Geobiology Gordon Research Conference, American Geophysical Union (AGU) Fall Meeting, Czech Society for Stable Isotopes Meeting
- 2021 International Arctic Workshop, European Geosciences Union (EGU) General Assembly, Rocky Mountain Geobiology Symposium, American Geophysical Union (AGU) Fall Meeting
- 2019 International Clumped Isotope Workshop, Rocky Mountain Geobiology Symposium, Front Range Microbiome Symposium, Astrobiology Science Conference (AbSciCon), Gordon Research Conference on Applied & Environmental Microbiology, Water Isotopes and Climate Workshop, American Geophysical Union (AGU) Fall Meeting
- 2018 Rocky Mountain Geobiology Symposium, Northeast Geobiology Symposium, Goldschmidt Conference, Front Range Isotope Day, Geological Society of America (GSA) Annual Meeting, American Geophysical Union (AGU) Fall Meeting
- 2017 Astrobiology Science Conference (AbSciCon), Geobiology Society Conference, Goldschmidt Conference, Geological Society of America (GSA) Annual Meeting
- 2016 American Society for Microbiology (ASM) Microbe Meeting, American Geophysical Union (AGU)